

Software Requirements Specification (SRS)

IoT Automated Smart Irrigation and Soil Moisture Analytics System

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Repository:	https://github.com/projectvault/smart-irrigation-iot.git
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1. Introduction & Scope

This Software Requirements Specification (SRS) details the functional, operational, and interface requirements for the IoT Smart Irrigation controller firmware and cloud monitoring dashboard.

2. Functional Requirements

- FR-01 (Sensor Data Acquisition): The ESP32 node must sample moisture and temperature sensors at 15-second intervals.
- FR-02 (Threshold Evaluation): The system must trigger solenoid valve relay GPIO pin HIGH when calibrated VWC drops below 35%.
- FR-03 (MQTT Broker Telemetry): Telemetry packets containing node_id, timestamp, vwc, and battery_voltage must publish to 'farm/zone1/telemetry'.
- FR-04 (Manual Override): The cloud dashboard must provide a secure switch to manually open or close water valves.
- FR-05 (Fail-Safe Timer): The valve must auto-close after 5 continuous minutes of pumping if sensor readings do not update.

3. Non-Functional Requirements

- Reliability: System must operate uninterrupted with 99.5% uptime during cultivation seasons.
- Power Consumption: Edge nodes must consume less than 80mA in active mode and 15uA in deep sleep.
- Response Latency: Valve activation commands must execute within 1.5 seconds of threshold trigger.

4. Hardware & Software Interfaces

- Microcontroller: ESP32 Dual-Core Tensilica Xtensa 32-bit LX6 @ 240MHz.
- Protocol: MQTT v3.1.1 over TLS 1.3 / TCP Port 8883.
- Database: PostgreSQL with TimescaleDB extension for time-series sensor points.